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Yushi Yang

AI Researcher · AI Engineer

LinkedIn
GitHub

Education

PhD in Social Data Science, University of Oxford [Website] [Scholar] [Lab] **Oct 2023 — Current**
- LLM Post-training, LLM Agents, Technical AI Safety; Reason with Machines Lab (led by Prof. Adam Mahdi)
- 3 first-author and 6 co-authored papers (NeurIPS, EMNLP; with Harvard); Department Scholarship for Exceptional Merit

MSc in Artificial Intelligence, Imperial College London **Oct 2021 — Sep 2022**
- Deep Learning, NLP, Reinforcement Learning; Software Engineering Group Project Prize (Partnered with The Trade Desk)

MSc in Statistical Science, University of Oxford **Oct 2020 — Sep 2021**
- Statistical Machine Learning, Bayesian Methods, Applied Statistics

BSc in Mathematics, Operational Research, Statistics, Economics (MORSE), University of Warwick **Oct 2017 — June 2020**
- Warwick Statistics Academic Excellence Prizes 2018 and 2019 (Top 1% and 3%)

Research Experience (AI Research)

LLM Post-training Researcher - Oxford Reasoning with Machines Lab [Code] **June 2024 — Current**
- Created a PyTorch DPO pipeline with 62% toxicity reduction on four LLMs and a neuron-editing method that replicates DPO effects
- Trained PPO and GRPO-based LLM search agents and developed attacks reducing answer safety by 83% via RL reward exploitation

Recommender System Developer - Oxford Human-Centred Computing Group [Code] **July 2023 — Sep 2023**
- Adapted the backend JavaScript for a movie-recommendation web app on Solid, Sir Tim Berners-Lee's decentralised data platform
- Integrated a collaborative filtering feature (Locality Sensitive Hashing) into the web app for privacy-preserving recommendations

Multimodal AI Researcher (MSc Project) - Department of Computing, Imperial College London [Code] **July 2022 — Sep 2022**
- Developed a TensorFlow concept extraction pipeline for a visual-language model to interpret the classified sentiments to artworks
- Processed 80k image-text pairs on Slurm GPUs to analyse how visual traits and text concepts jointly shape viewer emotions to art

Work Experience (AI Engineering)

AI Engineer (Part-time) - Reply **May 2025 — Current**
- Developed a multi-agent pipeline (LangGraph, Browse-use) to merge duplicate Salesforce records for a luxury automotive client
- Built an LLM-as-judge to extract agent decisions from logs with 90% accuracy and deployed the agents on AWS (Bedrock, Fargate)
- Architected a LangGraph multi-agent pipeline with routing, RAG and tool calling to answer HR and IT queries for an aviation client
- Constructed a 75k-document Ancient Greek benchmark with Mistral AI to evaluate fine-tuned LLMs on scholarly gap-filling quality

Data Analyst Intern - United Nations Development Programme (UNDP) **Nov 2022 — April 2023**
- Automated knowledge entity mapping in Microsoft Viva Topics using Power Automate to accelerate internal knowledge retrieval
- Built Power BI dashboards to analyse 10k access records to identify topic knowledge gaps for experts in Poverty & Inequality CoP

AI Engineer (MSc Project) - The Trade Desk & Imperial College London [Code] **Mar 2022 — May 2022**
- Built a BERT-based Python package for NER and sentiment analysis on news headlines and deployed with Docker on AWS Batch
- Processed 1M+ news headlines in 3 hours and built QuickSight dashboards to aid sentiment-driven ad bidding on global brands

Projects [GitHub]

- Built GPT-2 from scratch in PyTorch to generate Shakespearean text [Code]
- Classified patronising and condescending language with ensembled BERT models [Code]
- Predicted gun violence contagion rates in US cities using Hawkes processes with Bayesian inference [Code]

Skills

Programming	Python (PyTorch, PySpark, scikit-learn, Pandas, NumPy, Matplotlib), R (ggplot2, dplyr)
MLOps	AWS Certified Cloud Practitioner (Bedrock, SageMaker, QuickSight), LangGraph, Hugging Face, Git, MySQL
LLM	Post-training (LoRA, SFT, DPO, PPO, GRPO), RAG, Mechanistic interpretability
Models	Transformer, BERT, CNN, Autoencoders, GAN, XGBoost, Collaborative filtering, Clustering

PUBLICATIONS [GOOGLE SCHOLAR]

- How Does DPO Reduce Toxicity? A Mechanistic Neuron-Level Analysis [Paper] — **EMNLP 2025 Main** — *1st author*
- LLMs Don't Know Their Own Decision Boundaries: The Unreliability of Self-Generated Counterfactual Explanations [Paper] — **EMNLP 2025 Main** — *3rd author*
- Measuring What Matters: Construct Validity in Large Language Model Benchmarks [Paper] — **NeurIPS 2025** — *Co-author*
- Agentic Reinforcement Learning for Search Is Unsafe [Paper] — **ICLR 2026 (under review)** — *1st author*
- It's a TRAP! Task-Redirecting Agent Persuasion Benchmark for Web Agents — **ICLR 2026 (under review)** — *2nd author*
- Evaluating Fine-Tuning Efficiency of Human-Inspired Learning Strategies in Medical Question Answering [Paper] — **NeurIPS 2024 Workshop** — *1st author*
- Can Sparse Autoencoders Be Used to Decompose and Interpret Steering Vectors? [Paper] — **NeurIPS 2024 Workshop** — *2nd author*
- Robust LLM Unlearning with MUDMAN: Meta-Unlearning with Disruption Masking And Normalization [Paper] — **NeurIPS 2025 Workshop** — *2nd author*